Lecture 7(b): Flexibility quantification

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Agenda

- Motivation for flexibility quantification
- Proposed FQ
- Mathematical model
- Metrics for FQ

Motivation for flexibility quantification

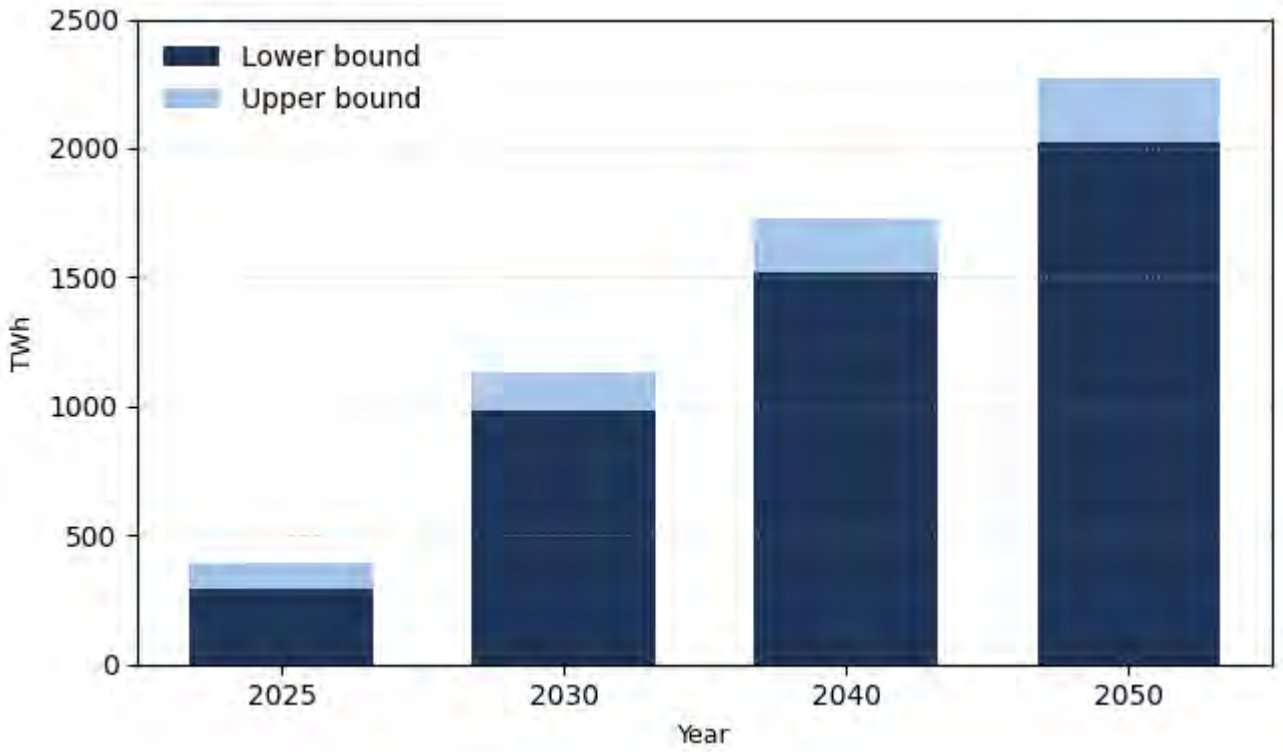
Context & Motivation for flexibility quantification

- Increasing share of DER in power system
- Pressure on distribution network: frequency variation, congestion,…
- Flexibility facilitates more efficient use of grid without need for expensive network upgrades

  → becomes valuable economic asset
- For owners of flexible assets → financial incentive in terms of energy cost savings and explicit revenues
- Flexibility over-estimation

Context & Motivation

- Increasing flexibility needs
- Elia: need for 6–7 GW of flexibility in the last hours before real time in Belgian system to comply with Loss of Load Expectation (LOLE) of 3 h/year [2]
- Adequate quantification is key to value it



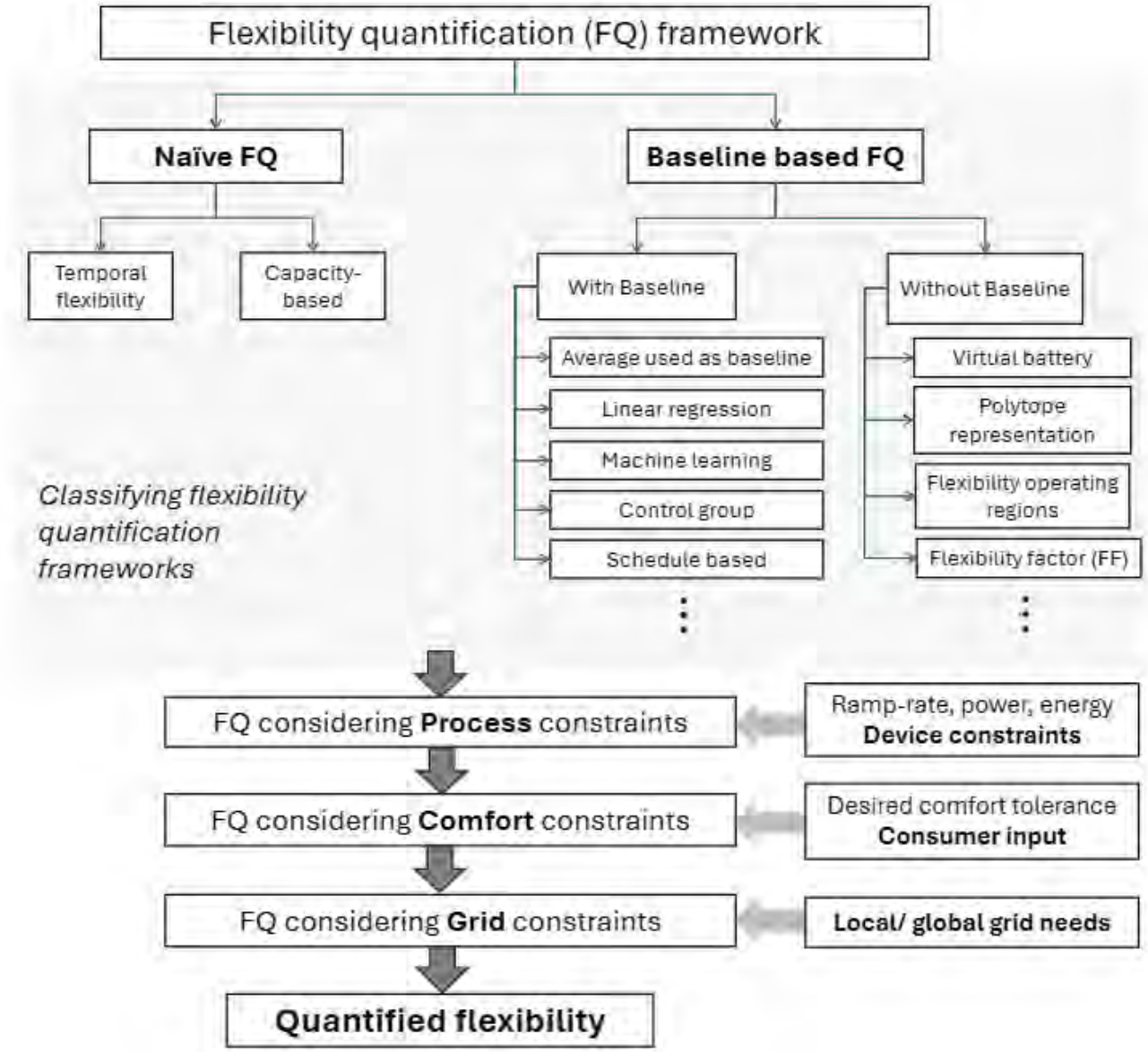
Projected daily (intraday) flexibility needs in European power system to maintain SoS.
(Image edited from Eurelectric study [1])

[1] Eurelectric and Compass Lexecon, “Redefining energy security in the age of electricity”, 2025.

[2] Elia, “Adequacy and flexibility study for Belgium (2026–2036)”, June 2025.

Context & Motivation

- Classification of existing frameworks
- Most existing flexibility quantification frameworks do not (i) respect process physics, (ii) capture the comfort constraints, and (iii) track explicit grid needs, simultaneously.
- This leads to an overestimation of available flexibility.



Problem Statement

- Develop a general flexibility quantification framework that integrates process constraints, comfort constraints and explicit grid-need signals.

  - Keep the implementation computationally tractable.

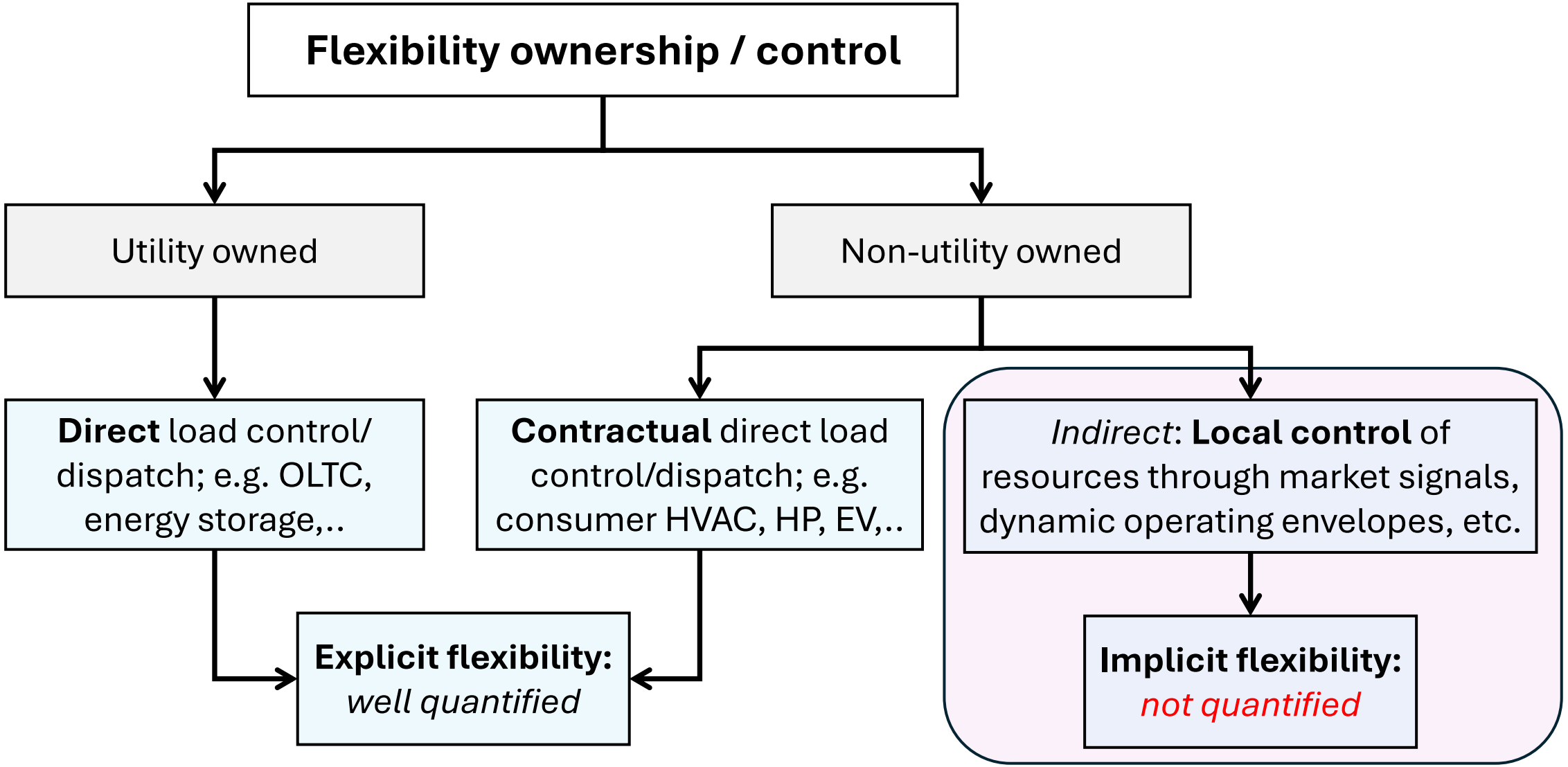
  - Apply the framework to three commonly used DERs:
    - electric vehicles (EV),
    - heat pumps (HP) and
    - inverter-interfaced photovoltaic generation (PV).

  - Perform sensitivity analyses to obtain insights in what drives flexibility for every DER.

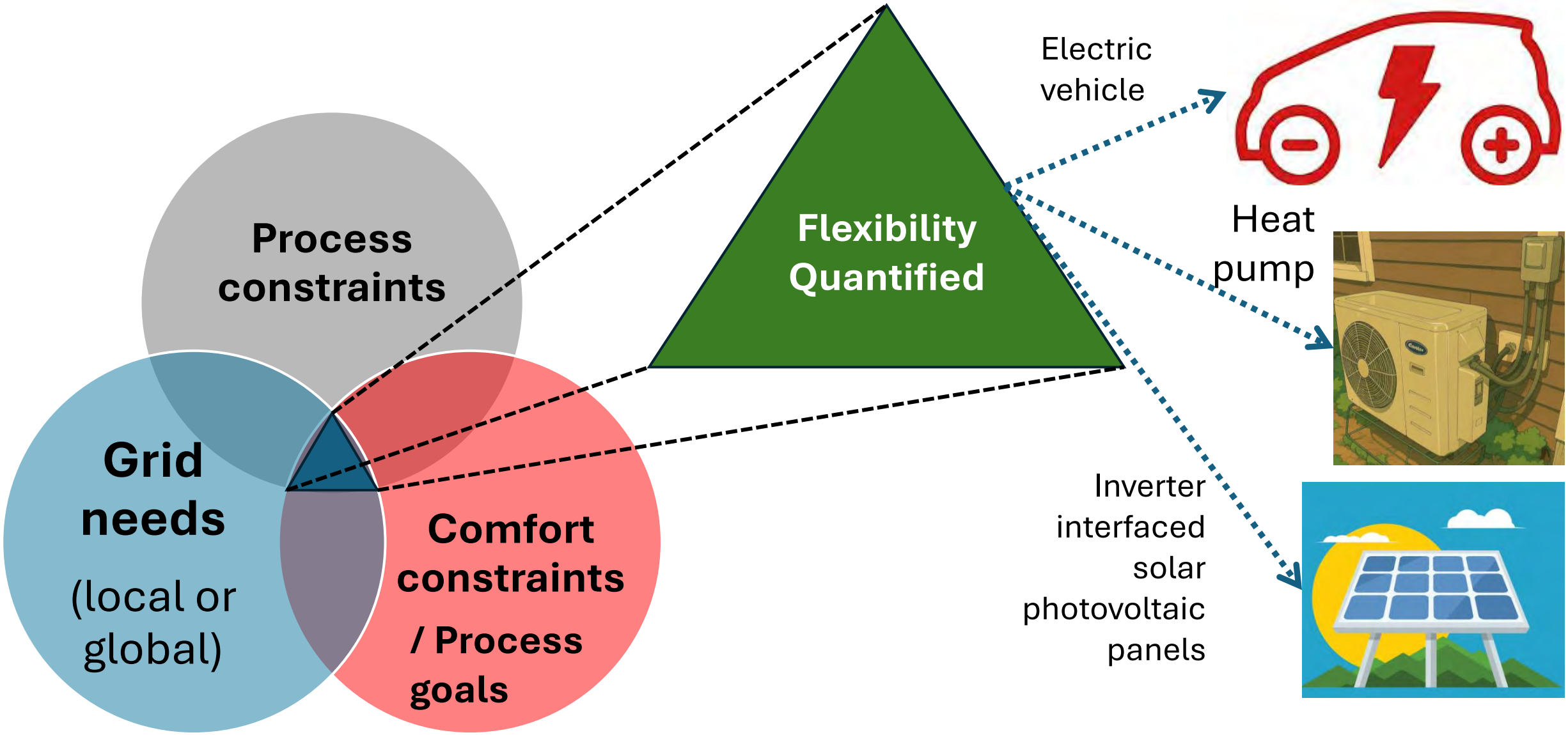
Proposed FQ

Revisiting classification

Positioning FQ



Proposed flexibility definition



What are process constraints?

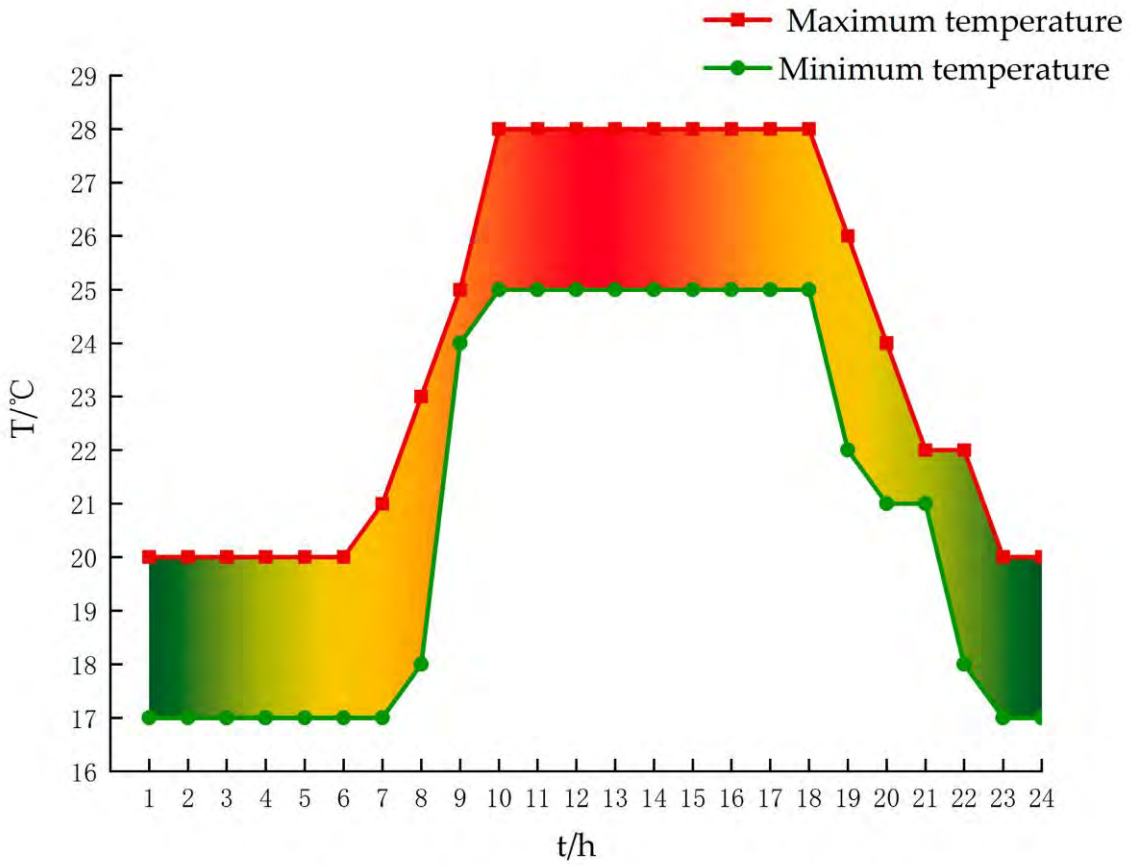
- Maximum, minimum power, ramp rate, energy constraints
- Connection constraint
- *For battery based DERs*: minimum state of charge limit
- *For thermostatic DERs*: Maximum heating power limit, safety temperature
- *For duty cycle based DERs*: Fixed modes of duty cycle dependent on pay load
- *For distributed generation based DER*: inverter limit, curtailment threshold

Thermostatic DERs include Heat pumps, electric water heaters, dryers, boilers, air conditioners, HVAC, space heaters

Duty cycle based DERs include dishwasher, washing machine, pool pumps

What are comfort constraints?

- Comfort constraint
- Consumer preference
- Deadline constraint
- *For distributed generation based DER*: Minimize curtailment



Bai, Yifan, et al. "A Hybrid Control Strategy for Multi-Timescale Air Conditioning Load Demand Response." *Smart Cities* 8.6 (2025): 204.

Handout

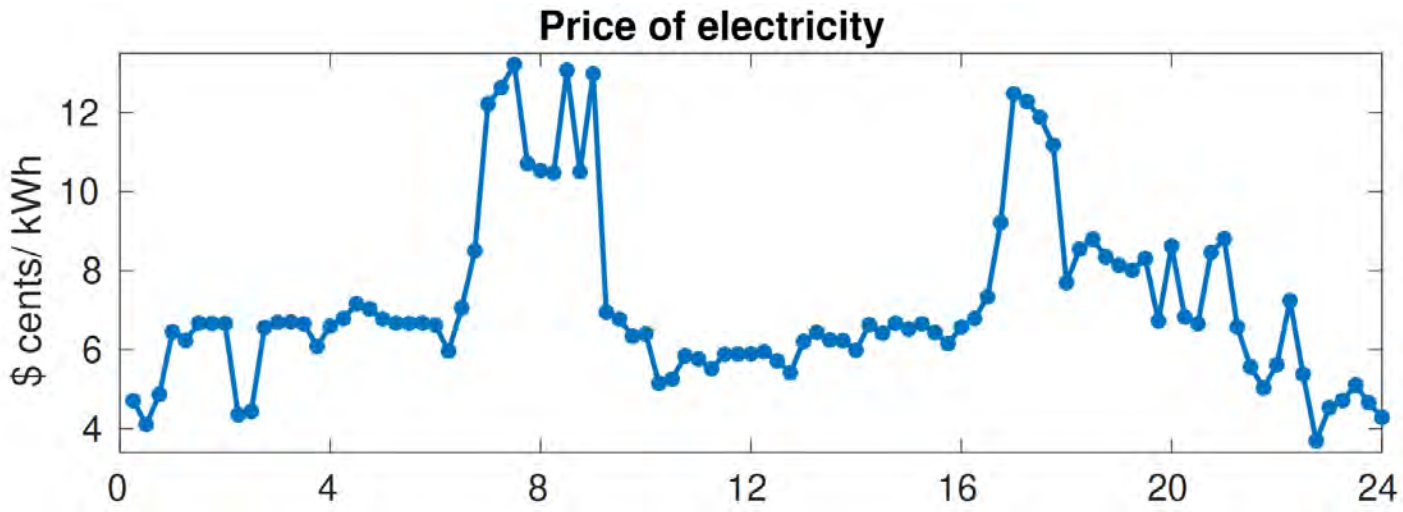
Handout: **Lecture 7**. DER process and comfort constraints:

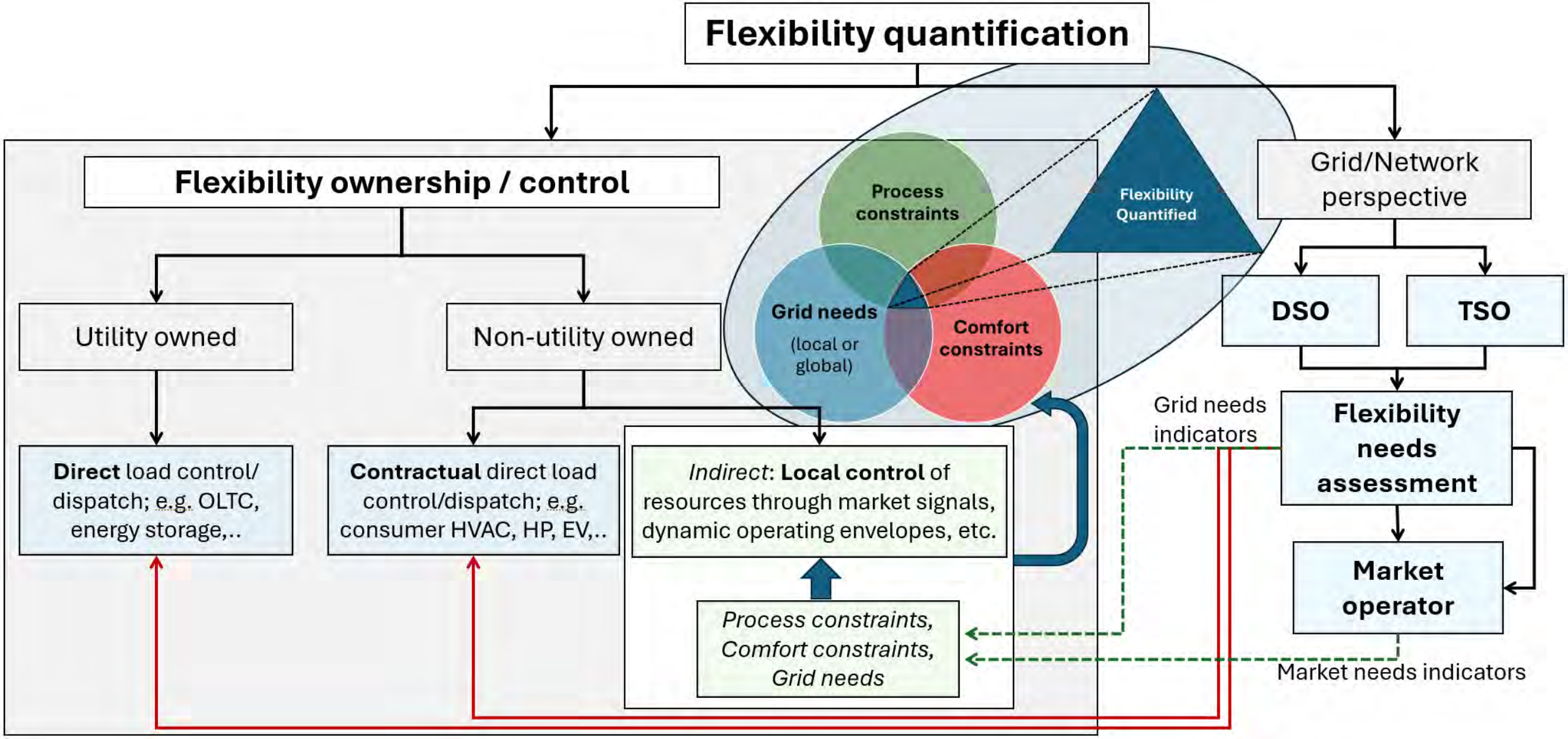
DER Type/Source	Process Constraints (Technical, Operational limits)	Comfort Constraints (User / Service Requirements: preset / dynamic)	Dynamic preference
Heat Pump (HP) (Space Heating/Cooling)	• Thermal dynamics of building (1R1C, 4R4C models). • Compressor cycling limits. • Max/min supply temperature. • Rated Coefficient of performance (COP) and defrost cycles.	• Maintain indoor temp within comfort band (e.g., 20–24°C). • Pre-heating/cooling must not degrade comfort.	Yes
Tumble Dryer	• Thermal load and cycle duration fixed.	• Dryness level and completion time / deadline constraints.	Yes
Electric Water Heater (EWH)	• Tank thermal inertia. • Max heating element power. • Safety temperature limit.	• Minimum hot water temperature (e.g., ≥50°C). • Adequate supply during peak use times (morning/evening).	Maybe
Tumble Dryer	• Thermal load and cycle duration fixed.	• Dryness level and completion time / deadline constraints.	Yes
Electric Boilers	• Maximum heating power and temperature ramp limits.	• Minimum outlet water temperature for comfort band.	Yes
Air Conditioners	• Compressor duty cycle. • Cooling capacity limits.	• Indoor temperature band constraints; humidity preferences.	Maybe
Space Heaters [1] (Resistive, infrared)	• Power rating fixed; no thermal storage.	• Indoor temperature must remain in comfort band.	Maybe
Electric Vehicle (EV)	• Battery charging curve (CC-CV). • Max charging power & connector rating. • Battery health/SOC limits.	• Required SOC by departure time (deadline constraint). • User preference for minimum SOC (range anxiety).	Yes
Home Battery Storage	• State of charge (SOC) bounds; charge/discharge rates; round-trip efficiency.	• User preference for backup SOC reserve. • Limiting cycles. • cost minimization. • Target SOC (example of deadline constraint).	Yes
Solar Photovoltaic (PV) → distributed generation-based DER	• Irradiance-driven generation. • Inverter rating limits (P/Q). • Curtailment thresholds.	• Minimize curtailment, • Optional: user preference to maximize self-consumption.	No

What are grid needs?

Grid needs signal (GNS) can be

- Direct indicators:
  - Congestion signal
  - Imbalance indicators
  - Centralized / decentralized
- Indirect indicators:
  - Market signals
    - Prices signal



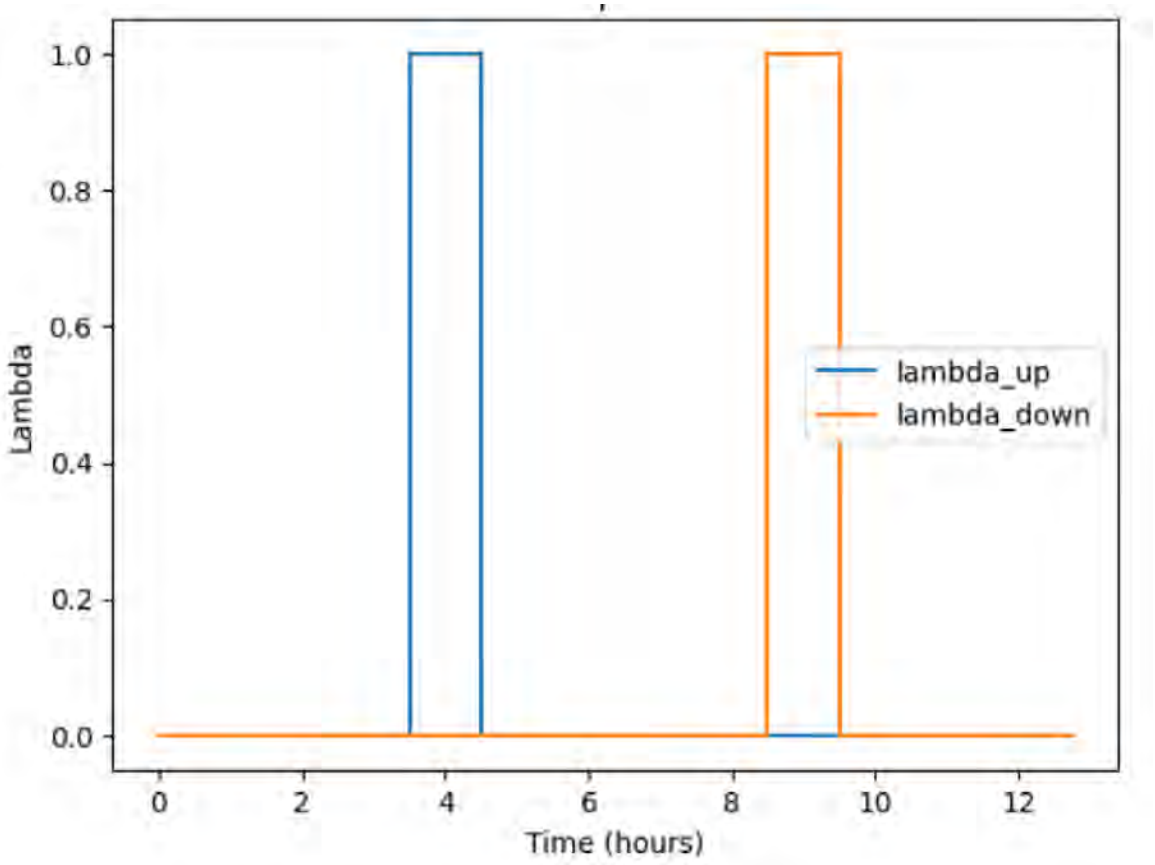

Flexibility quantification
Flexibility ownership / control
Utility owned
Non-utility owned
Direct load control/ dispatch; e.g. OLTC, energy storage,..
Contractual direct load control/dispatch; e.g. consumer HVAC, HP, EV,..
Indirect: Local control of resources through market signals, dynamic operating envelopes, etc.
Process constraints, Comfort constraints, Grid needs
Process constraints
Grid needs
(local or global)
Comfort constraints
Flexibility Quantified
Grid/Network perspective
DSO
TSO
Flexibility needs assessment
Market operator
Grid needs indicators
Market needs indicators

Mathematical model for FQ

Generalized grid needs signal

- $\lambda_{up} = 1$ → grid needing less generation or more consumption

- $\lambda_{down} = 1$ → grid needing more generation or less consumption



Device constraints

- Denoting flexibility:

$$P_t^{\text{flex}} = P_t^{\text{new}} - P_t^{\text{baseline}}$$

- Device constraints:

Power limit: $P^{\min} \leq P_t^{\text{new}} \leq P^{\max}, \quad \forall t$

Ramp rate: $|P_t^{\text{new}} - P_{t-1}^{\text{new}}| \leq \Delta_P^{\max},$

Connection limit:

$$-P^{\max,\text{connection}} \leq P_t^{\text{inelastic}} + P_t^{\text{new}} \leq P^{\max,\text{connection}}$$

Objective function

- Maximizing profit

$$\max_{P_t^{\text{new}}} \Delta t \cdot \sum_{t=1}^{T} \left(\lambda_t^{\text{up}} \cdot [P_t^{\text{flex}}]^+ + \lambda_t^{\text{down}} \cdot [-P_t^{\text{flex}}]^+ \right)$$

- $[k]^+$ denotes max(0, k)

Reformulating the objective function

$$\delta_t^+ = \max\{P^{\text{flex}}, 0\}, \qquad \delta_t^- = \max\{-P_t^{\text{flex}}, 0\}, \quad \forall\, t$$

$$\max \quad \Delta t \sum_{t=1}^{T} \left(\lambda_t^{\text{up}}\, \delta_t^+ + \lambda_t^{\text{down}}\, \delta_t^- \right) \quad \delta_t^+,\ \delta_t^- \geq 0$$

$$\delta_t^+ \ \leq\ M_1\, b_t$$

$$\delta_t^- \ \leq\ M_1\, (1 - b_t)$$

FQ model:

Objective function →

$$\max \quad \Delta t \sum_{t=1}^{T} \left(\lambda_t^{\text{up}} \, \delta_t^{+} + \lambda_t^{\text{down}} \, \delta_t^{-} \right)$$

Constraint →

Power limit: $P^{\min} \leq P_t^{\text{new}} \leq P^{\max}, \quad \forall t$

Ramp rate: $\left| P_t^{\text{new}} - P_{t-1}^{\text{new}} \right| \leq \Delta_P^{\max},$

Connection limit:

$$-P^{\max,\text{connection}} \leq P_t^{\text{inelastic}} + P_t^{\text{new}} \leq P^{\max,\text{connection}}$$

$$\delta_t^{+} \leq M_1 \, b_t$$

$$\delta_t^{-} \leq M_1 \, (1 - b_t)$$

$$\delta_t^{+}, \, \delta_t^{-} \geq 0$$

+

DER comfort constraint

Metrics for FQ

Key performance indicators

- Quantified flexibility energy: The FQ in terms of energy is denoted as

$$E^{\text{flex}} = \Delta t \sum_{t=1}^{T} \left(\delta_t^{+} \, \mathbb{1}(\lambda_t^{\text{up}} > 0) + \delta_t^{-} \, \mathbb{1}(\lambda_t^{\text{down}} > 0) \right)$$

$$E_{\text{up}}^{\text{flex}} = \Delta t \sum_{t=1}^{T} \delta_t^{+} \, \mathbb{1}(\lambda_t^{\text{up}} > 0) \qquad E_{\text{down}}^{\text{flex}} = \Delta t \sum_{t=1}^{T} \delta_t^{-} \, \mathbb{1}(\lambda_t^{\text{down}} > 0)$$

$$E_{\text{up}}^{\text{flex}} \oplus E_{\text{down}}^{\text{flex}}$$

Key performance indicators

Total energy throughput ratio: Assets that provide flexibility may need to consume more electrical energy than in the baseline/ nominal case without flexibility considerations. This metric is crucial for getting insights into additional energy consumption costs

$$\epsilon_{\text{needed}}^{\text{abs}} = \frac{E^{\text{abs}}}{E^{\text{needed}}}$$

$$E^{\text{abs}} = \sum_t \left| P_t^{\text{new},\star} \right| \Delta t \qquad E^{\text{needed}} = \sum_t \left| P_t^{\text{baseline}} \right| \Delta t$$

Factor is greater than or equal to 1

Key performance indicators

Fraction of flexibility harvested:

$$\epsilon_{\text{needed}}^{\text{flex}} = \frac{E^{\text{flex}}}{E^{\text{needed}}}$$

Conclusions

- Need for accurate flexibility estimation
- A new method for quantification of flexibility to account for
  - Process constraints
  - Comfort constraints
  - Grid needs consideration
- Mathematical model for flexibility quantification
- Key performance indicators for FQ

Course acknowledgements

Kennis kent geen einde

IMPROCAP

https://www.improcap.eu/

DiGiRES

https://www.digi-res.eu/

C2-FlexIQ

https://research.kuleuven.be/portal/nl/project/3E240512

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Thank you for your attention!

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